

Fully Interconnected Die-to-Wafer Hybrid Copper Bonding Process for Opto-Electrical Routing Platforms

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Abstract We demonstrate simultaneous opto-electrical interconnection on a die-to-wafer hybrid bonding system. CMP erosion is $< 100 \text{ \AA}$ even with pattern density mismatch between optic and electric layers. Measured evanescent coupler loss is 0.32 dB at 1310 nm and measured electrical resistance is 0.30 Ω . ©2026 The Author(s)

Introduction

The rapid growth of artificial intelligence (AI), high-performance computing (HPC), and hyper-scale data centers is driving an unprecedented demand for high-performance semiconductor devices and advanced packaging architectures [1].

In recent industry trends, chiplet-based heterogeneous packaging processes are increasingly adopted to obtain various advantages such as overcoming the limitations of monolithic scaling, enabling process separation by chip functionality, achieving highly scalable architectures, and improving interconnection bandwidth while reducing power consumption. Interposer-based chip-to-chip integration (2.xD), 3D stacked chips with TSV approaches, and hybrid bonding are representative chip or wafer level package structures for this purpose [2]. These architectures rely heavily on high-speed electrical interconnects. However, as data rates scale beyond 100-200 Gb/s per lane with roadmaps approaching 400Gb/s, purely electrical interconnect system faces increasing challenges, including signal attenuation,

noise, and power dissipation [3]. These challenges are further exacerbated as advanced packaging expands its scope from chip-level to wafer- and panel-level interconnections to implement large-scale packaging architectures like system-on-wafer and system-on-panel. The primary benefit of such a large scale system comes from increased bandwidth through high density wafer-level interconnect fabric. However, this long range routing, which is essential for large-scale system, intensely suffers from low signal integrity, transmission quality degradation and limited I/O bandwidth in high-speed data processing at 200Gb/s and beyond [1]. To overcome these limitations, optical routing and signalling have emerged as attractive solutions, offering low propagation loss and high-bandwidth data transfer over wide range of routing traces. In particular, hybrid bonding based optical interposers employing evanescent couplers (EVC) have gained significant attention. Unlike conventional macro-

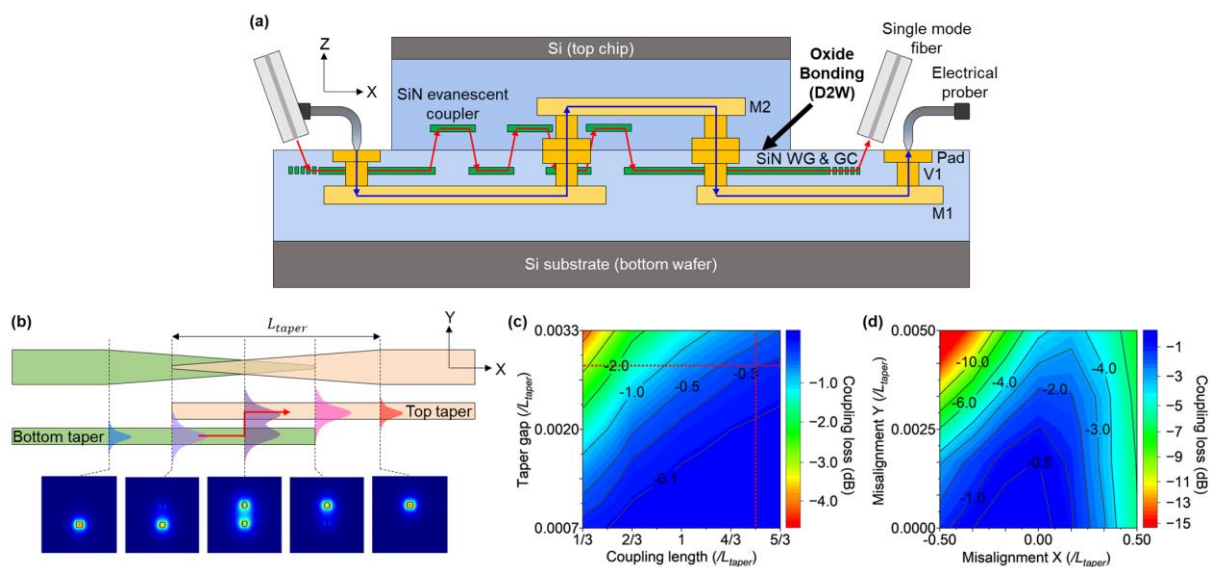


Fig. 1: (a) Schematic of D2W hybrid bonding structure. Optical connection is achieved using a SiN evanescent coupler, while Cu pads provide electrical interconnection. (b) Linear tapered SiN evanescent coupler and adiabatic mode transition. (c)–(d) Coupling efficiency as a function of misalignment and geometrical parameters. The convergence of the red dashed line represents the optical loss at target coupling length and taper gap.

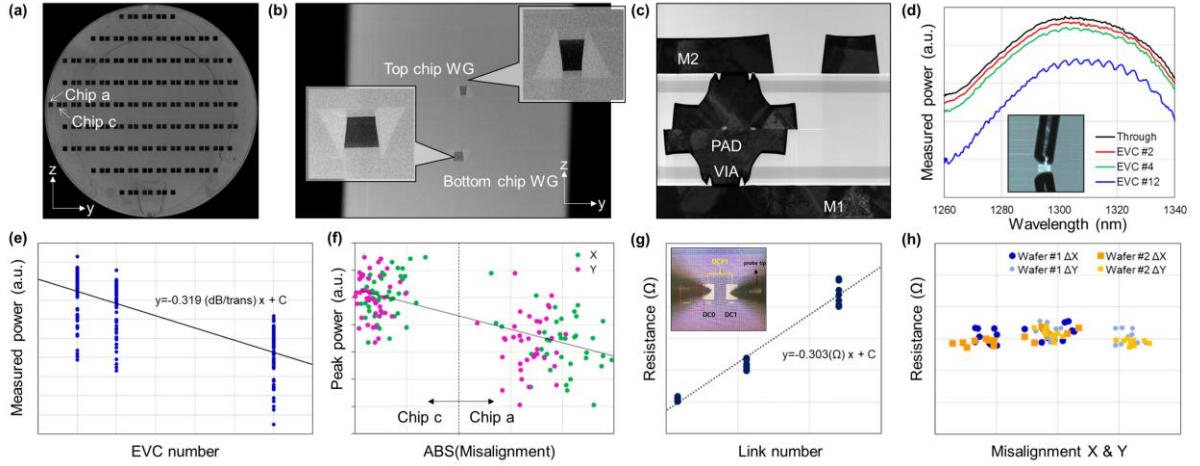


Fig. 2: (a) Scanning acoustic tomography of D2W bonding wafer. Two chips (a and c) are attached on two different locations within a field. (b) Cross section view of the top and bottom EVC (c) Cross section view of electrical interconnection between Cu pads. (d) Transmission spectra of evanescent couplers according to transition number. (e) Transition loss of the EVC obtained through linear regression. The measured loss is 0.319 dB per transition (f) Peak power reduction as misalignment increases. (g) Measured electrical resistance per a link between M1 to M2 through cutback structures. The measured resistance is 0.303 Ω per link (h) Electrical resistance depending on misalignment.

scale optical couplers such as optical fibers or glass lenses, EVCs operate at the nano-micro scale and have a wide optical bandwidth and low optical loss, making them highly suitable for on-chip and on-package integration [4], [5]. In this aspect, hybrid bonding platform provides an ideal combination of optical and electrical co-integration. The process technology for hybrid bonding offers precise alignment accuracy of less than 1 μm , and enables direct surface attachment between chips without any intervening material in optical path. The ultra-high-density Cu-Cu bonding can supplement short-reach electrical interconnections, and also provide I/O and power delivery path [6]. Therefore, the simultaneous introduction of hybrid bonding and evanescent coupler technologies is important for the introduction of next-generation heterogeneous integration platform.

In this paper, we demonstrate a die-to-wafer (D2W) hybrid copper bonding process incorporating a SiN evanescent coupler. Simultaneous optical and electrical interconnection is successfully achieved. We expect that the proposed hybrid bonding process provides a practical and scalable solution for heterogeneously integrated package architectures.

Design and Fabrication

To enable simultaneous optical and electrical interconnection, SiN-based EVC and Cu daisy-chain with hybrid bonding structures are designed and implemented on both the top die and bottom 12-inch wafer (Fig.1 (a)). The top and bottom Cu pads are connected via direct Cu-Cu bonding to achieve low-resistance electrical pathways. In parallel, optical coupling between the

SiN EVCs are realized through direct oxide-to-oxide bonding to ensure minimal optical loss across the interfaces. To minimize coupling loss arising from complex fabrication processes, a simple linear tapered EVC is employed, featuring waveguide width of L_{taper} . Coupling length is deliberately selected to compensate for optical loss by large oxide gap, which is required to accommodate Cu pad layer fabrication. Under this condition, the decay length of the evanescent field is comparable or shorter than the oxide gap thickness, which breaks adiabatic mode transition condition and leads to significant coupling loss. With the given oxide gap and coupling length, a coupling loss of 0.314 dB is obtained for the EVC, as evaluated 3D FDTD simulation. The alignment tolerance of EVC is also checked through FDTD simulation, a 1 dB coupling loss tolerance is achieved under a misalignment of 1 μm along the X and Y axes. This tolerance margin is sufficient to be accommodated by hybrid bonding process (Fig.1 (b)~(d)). In the chip fabrication process, a thick bottom oxide layer is deposited to suppress optical dissipation into Si substrate, caused by large mode field diameter of tapered waveguide. Subsequently, Cu M1 layer is patterned, followed by thin oxide buffer layer. Next, a SiN layer is patterned to make grating for optical I/O, waveguides and evanescent coupler., Cu pads and vias are fabricated on top of SiN layer for hybrid bonding. Finally, hybrid Cu bonding is performed between 12-inch wafer and mm-scale chips.

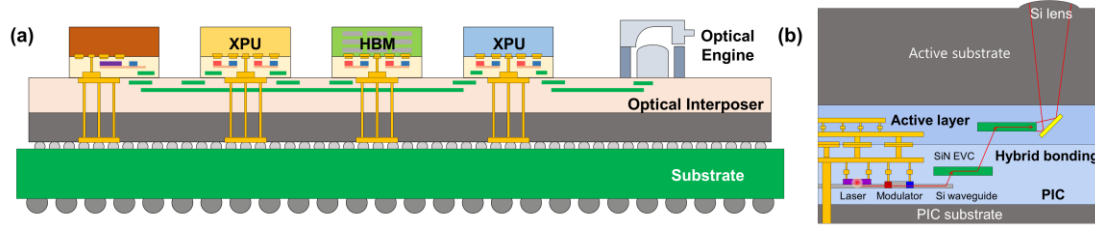


Fig. 3: (a) Optical interposer with heterogeneous integration featuring a high-performance chip and optical engine. Reprinted from IMEC [7], [8] (b) Example of a hybrid bonding based co-packaged optic platform.

Experimental Results

We experimentally implemented the photonic chip-to-wafer hybrid bonding. **Fig.2 (a)** is a scanning acoustic tomography of post-bond wafer, showing the absence of voids at the bonding interface. This is attributed to the excellent erosion of less than 100 Å per 100 μm square area after Cu pad CMP, despite the large pattern density inhomogeneity between uniform Cu pads domain and the SiN photonic domain, where large metal keep-out-zone(KOZ) is required to minimize coupling interference. **Fig.2 (b) and (c)** show cross-sectional TEM images of the evanescent coupler and bonded Cu pads. Through precise alignment, a robust submicron misalignment is obtained along both X and Y axes, which are well below the 1 dB loss tolerance range predicted by FDTD simulations. The photonic circuits consist of butt-coupled SiN grating I/Os, waveguides, and linear tapered EVCs. These insertion losses are extracted using linear regression analysis based on cutback structures with increasing EVC transitions. **Fig.2 (d)** shows the measured power spectra of the waveguide-through and EVC cutback structures. Under high-quality hybrid bonding conditions, spectral distortion rarely occur as light propagates through the EVC. Notably, the insertion loss of EVC is measured to be 0.319 dB per transition at 1310 nm (**Fig.2 (e)**). In contrast, as the misalignment increases, it leads a reduction in peak transmitted power due to increased insertion loss in the EVC, consistent with FDTD simulation predictions (see **Fig.2 (f)**). chip-A and chip-C are compared in **Fig.2 (a)**). In parallel, electrical characterization is performed to verify the interconnectivity between M1 and M2 circuits using cutback daisy-chain structures. The measured resistance is 0.303 Ω per link, which is consistent with typical values (**Fig.2 (h)**). Finally, the resistance remains nearly unchanged for the same misalignment levels (**Fig.2 (i)**), showing less sensitivity than EVC measurement.

Conclusion and Summary

In summary, we successfully demonstrate hybrid bonding between photonic chips and 12-inch

optical interposer. High bonding quality is achieved with excellent CMP erosion (<100 Å per 100 μm square area after CMP) and precise alignment (<1 μm). Under this condition, EVC loss is measured to be 0.319 dB/transition at 1310 nm. On the other hand, daisy chain resistance is measured to be 0.303 Ω. The proposed hybrid bonding system is directly applicable to optical/electrical routing platforms such as package- or wafer-level heterogeneous interposers [6], [8] and co-packaged optics, enabling low-loss optical and high-density electrical interconnections (see **Fig.3**).

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